

COGNIZANT DIGITAL NURTURE

Java Full Stack Program (Java FSE)

WEEK 1 REPORT

Core Backend Foundations & Best Practices

Topics Covered:

1. Design Patterns & Principles (11 Exercises)
2. Data Structures & Algorithms (7 Exercises)
3. PL/SQL Programming (7 Exercises)
4. JUnit & Mockito Testing (68 Unit/Integration Tests)
5. SLF4J & Logback Logging (Console & File Appenders)

Submitted by:

Ishan Parmar

Digital Nurture Candidate

Table of Contents

1. Design Patterns & Principles Overview	Page 3
- Key Implementations (Singleton, Factory, Builder, etc.)	Page 3
- Program Execution Outputs	Page 6
2. Data Structures & Algorithms Overview	Page 7
- Inventory, E-commerce, Sorting, Employee, Tasks, Search, Forecasting	Page 7
- Performance & Time Complexity Analysis Outputs	Page 9
3. PL/SQL Database Programming Scripts	Page 11
- Schema Definitions, Procedures, Functions, Triggers, Packages	Page 11
4. JUnit & Mockito Testing Framework	Page 14
- Calculator, Order, Exception, Timeout, Mock Stubbing, Spring MVC/JPA Tests	Page 14
- 68 Tests Run Execution Outputs	Page 17
5. SLF4J & Logback Logging Project	Page 19
- Appenders & Custom Configurations (Console & app.log)	Page 19
- Logging Output File Records	Page 21

1. Design Patterns & Principles

Design patterns represent best practices used by experienced object-oriented software developers. In this section, we implement 11 distinct patterns representing Creational, Structural, and Behavioral patterns:

- Singleton Pattern: Logger class for double-checked lazy initialization.
- Factory Method Pattern: Document creator for Word, PDF, and Excel factories.
- Builder Pattern: Step-by-step assembly of complex Computer items.
- Adapter Pattern: Harmonizing PayPal, Stripe, and Square interfaces.
- Decorator Pattern: Notification wrappers extending Email with SMS and Slack.
- Proxy Pattern: Caching remote images with lazy initialization.
- Observer Pattern: Stock market updates pushed in real-time to active sessions.
- Strategy Pattern: Swapping Credit Card, PayPal, and Bitcoin algorithms.
- Command Pattern: Light controls bound to home commands.
- MVC Pattern: Student details partitioned into Model, View, and Controller.
- Dependency Injection: Injecting Customer repository objects into business services.

SingletonPatternExample.java

```
/*
 * Exercise 1: Implementing the Singleton Pattern
 *
 * Scenario: Ensure that a logging utility class has only one instance
 * throughout the application lifecycle to ensure consistent logging.
 *
 * The Singleton Pattern restricts the instantiation of a class to a single instance
 * and provides a global point of access to that instance. This is useful when exactl
y
 * one object is needed to coordinate actions across the system.
 */

// Logger class implementing the Singleton Pattern
class Logger {
    // Private static instance of the Logger class
    private static Logger instance;

    // Private constructor to prevent instantiation from outside
    private Logger() {
        System.out.println("Logger instance created.");
    }

    // Public static method to get the single instance of Logger (Thread-safe with la
zy initialization)
    public static synchronized Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    // Method to log messages
    public void log(String message) {
        System.out.println("[LOG] " + message);
    }
}
```

```

}

// Test class to verify Singleton implementation
public class SingletonPatternExample {
    public static void main(String[] args) {
        System.out.println("=== Singleton Pattern Example ===\n");

        // Get the first instance of Logger
        Logger logger1 = Logger.getInstance();
        logger1.log("This is the first log message.");

        // Get the second instance of Logger
        Logger logger2 = Logger.getInstance();
        logger2.log("This is the second log message.");

        // Verify both references point to the same instance
        System.out.println("\nlogger1 hashCode: " + logger1.hashCode());
        System.out.println("logger2 hashCode: " + logger2.hashCode());
        System.out.println("Are both instances the same? " + (logger1 == logger2));
    }
}

```

FactoryMethodPatternExample.java

```

/*
 * Exercise 2: Implementing the Factory Method Pattern
 *
 * Scenario: Developing a document management system that needs to create
 * different types of documents (Word, PDF, Excel).
 *
 * The Factory Method Pattern defines an interface for creating an object,
 * but lets subclasses alter the type of objects that will be created.
 */

// Abstract Document interface
abstract class Document {
    public abstract void open();
    public abstract void save();
    public abstract void close();
}

// Concrete Document: WordDocument
class WordDocument extends Document {
    @Override
    public void open() {
        System.out.println("Opening Word Document...");
    }

    @Override
    public void save() {
        System.out.println("Saving Word Document...");
    }

    @Override
    public void close() {

```

```
        System.out.println("Closing Word Document...");
    }
}

// Concrete Document: PdfDocument
class PdfDocument extends Document {
    @Override
    public void open() {
        System.out.println("Opening PDF Document...");
    }

    @Override
    public void save() {
        System.out.println("Saving PDF Document...");
    }

    @Override
    public void close() {
        System.out.println("Closing PDF Document...");
    }
}

// Concrete Document: ExcelDocument
class ExcelDocument extends Document {
    @Override
    public void open() {
        System.out.println("Opening Excel Document...");
    }

    @Override
    public void save() {
        System.out.println("Saving Excel Document...");
    }

    @Override
    public void close() {
        System.out.println("Closing Excel Document...");
    }
}

// Abstract Factory class
abstract class DocumentFactory {
    public abstract Document createDocument();
}

// Concrete Factory: WordDocumentFactory
class WordDocumentFactory extends DocumentFactory {
    @Override
    public Document createDocument() {
        return new WordDocument();
    }
}

// Concrete Factory: PdfDocumentFactory
class PdfDocumentFactory extends DocumentFactory {
    @Override
```

```

    public Document createDocument() {
        return new PdfDocument();
    }
}

// Concrete Factory: ExcelDocumentFactory
class ExcelDocumentFactory extends DocumentFactory {
    @Override
    public Document createDocument() {
        return new ExcelDocument();
    }
}

// Test class
public class FactoryMethodPatternExample {
    public static void main(String[] args) {
        System.out.println("=== Factory Method Pattern Example ===\n");

        // Create Word Document using Factory
        DocumentFactory wordFactory = new WordDocumentFactory();
        Document wordDoc = wordFactory.createDocument();
        wordDoc.open();
        wordDoc.save();
        wordDoc.close();

        System.out.println();

        // Create PDF Document using Factory
        DocumentFactory pdfFactory = new PdfDocumentFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();
        pdfDoc.save();
        pdfDoc.close();

        System.out.println();

        // Create Excel Document using Factory
        DocumentFactory excelFactory = new ExcelDocumentFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
        excelDoc.save();
        excelDoc.close();
    }
}

```

Design Patterns Execution Log (output.txt)

```

=== Singleton Pattern ===
=== Singleton Pattern Example ===

Logger instance created.
[LOG] This is the first log message.
[LOG] This is the second log message.

logger1 hashCode: 21685669

```

```
logger2 hashCode: 21685669
Are both instances the same? true
```

```
=== Factory Method Pattern ===
=== Factory Method Pattern Example ===
```

```
Opening Word Document...
Saving Word Document...
Closing Word Document...
```

```
Opening PDF Document...
Saving PDF Document...
Closing PDF Document...
```

```
Opening Excel Document...
Saving Excel Document...
Closing Excel Document...
```

```
=== Builder Pattern ===
=== Builder Pattern Example ===
```

```
--- Office PC ---
Computer Configuration:
  CPU: Intel i5
  RAM: 8GB
  Storage: 512GB SSD
  GPU: Integrated
  OS: Windows 11
  Bluetooth: Disabled
  WiFi: Enabled
```

```
--- Gaming PC ---
Computer Configuration:
  CPU: Intel i9
  RAM: 32GB
  Storage: 2TB NVMe SSD
  GPU: NVIDIA RTX 4090
  OS: Windows 11 Pro
  Bluetooth: Enabled
  WiFi: Enabled
```

```
--- Server ---
Computer Configuration:
  CPU: AMD EPYC
  RAM: 128GB
  Storage: 4TB NVMe RAID
  GPU: Integrated
  OS: Ubuntu Server 22.04
  Bluetooth: Disabled
  WiFi: Disabled
```

```
=== Adapter Pattern ===
=== Adapter Pattern Example ===
```

```
Processing payment of $100.0 through PayPal.
Processing payment of $250.5 through Stripe.
```

Processing payment of \$75.25 through Square.

All payment gateways processed through a unified interface!

=== Decorator Pattern ===

=== Decorator Pattern Example ===

--- Email Only ---

[Email] Sending: Hello, this is an email notification!

--- Email + SMS ---

[Email] Sending: Hello, this is a multi-channel notification!

[SMS] Sending: Hello, this is a multi-channel notification!

--- Email + SMS + Slack ---

[Email] Sending: Hello, this goes to all channels!

[SMS] Sending: Hello, this goes to all channels!

[Slack] Sending: Hello, this goes to all channels!

=== Proxy Pattern ===

=== Proxy Pattern Example ===

Images created but not yet loaded.

--- First access to image1 ---

ProxyImage: First access - loading image...

Loading image from remote server: photo1.jpg

Image loaded successfully: photo1.jpg

Displaying image: photo1.jpg

--- Second access to image1 (cached) ---

ProxyImage: Using cached image...

Displaying image: photo1.jpg

--- First access to image2 ---

ProxyImage: First access - loading image...

Loading image from remote server: photo2.jpg

Image loaded successfully: photo2.jpg

Displaying image: photo2.jpg

=== Observer Pattern ===

=== Observer Pattern Example ===

Observer registered for GOOGL

Observer registered for GOOGL

Observer registered for GOOGL

GOOGL price changed: \$2800.0 -> \$2850.5

[StockTracker - Mobile] GOOGL is now \$2850.5

[StockDashboard - Web] GOOGL is now \$2850.5

[InvestorApp - Mobile] GOOGL is now \$2850.5

GOOGL price changed: \$2850.5 -> \$2900.75

[StockTracker - Mobile] GOOGL is now \$2900.75

[StockDashboard - Web] GOOGL is now \$2900.75

[InvestorApp - Mobile] GOOGL is now \$2900.75

Observer deregistered from G00GL

G00GL price changed: \$2900.75 -> \$2875.0
[StockTracker - Mobile] G00GL is now \$2875.0
[StockDashboard - Web] G00GL is now \$2875.0

=== Strategy Pattern ===
=== Strategy Pattern Example ===

Paid \$150.0 using Credit Card ending in 1234
Paid \$89.99 using PayPal account: user@example.com
Paid \$500.0 using Bitcoin wallet: 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa

=== Command Pattern ===
=== Command Pattern Example ===

Remote button pressed -> Light is ON
Remote button pressed -> Light is OFF

=== MVC Pattern ===
=== MVC Pattern Example ===

Initial Student Details:
=== Student Details ===
Name: Ishan Parmar
ID: S001
Grade: A
=====

Updating student details...

Updated Student Details:
=== Student Details ===
Name: Ishan P.
ID: S001
Grade: A+
=====

=== Dependency Injection ===
=== Dependency Injection Example ===

Customer found: John Doe (ID: 1)
Customer found: Jane Smith (ID: 2)
Customer found: Ishan Parmar (ID: 3)
Customer with ID 4 not found.

2. Data Structures & Algorithms

Analyzing algorithm performance is key for high-load systems. We implement 7 exercises:

- Inventory Management: HashMap lookup ($O(1)$) vs ArrayList iteration.
- E-commerce Search: Linear Search $O(n)$ vs Binary Search $O(\log n)$.
- Sorting Customer Orders: Bubble Sort $O(n^2)$ vs Quick Sort $O(n \log n)$.
- Employee System: Shifting array deletion ($O(n)$) vs insertion.
- Task LinkedList: Custom Singly Linked List insertion and deletion.
- Library Search: Dynamic lookup comparison on sorted book items.
- Financial Forecasting: Call-stack recursive execution vs $O(1)$ space iterative design.

InventoryManagement.java

```
/*
 * Exercise 1: Inventory Management System
 *
 * Scenario: Efficient data storage and retrieval for a warehouse inventory.
 *
 * Why data structures are essential:
 * - Large inventories require efficient lookups, insertions, and deletions.
 * - HashMap provides  $O(1)$  average-case for get/put operations vs  $O(n)$  for ArrayList
search.
 * - Choosing the right data structure reduces time complexity significantly.
 *
 * Data structure chosen: HashMap (productId as key for  $O(1)$  lookups)
 *
 * Time Complexity Analysis:
 * - Add:  $O(1)$  average (HashMap.put)
 * - Update:  $O(1)$  average (HashMap.put after get)
 * - Delete:  $O(1)$  average (HashMap.remove)
 * - Search:  $O(1)$  average (HashMap.get)
 *
 * Optimization: HashMap provides the best average-case complexity.
 * For ordered traversal, TreeMap ( $O(\log n)$ ) could be used instead.
 */

import java.util.HashMap;
import java.util.Map;

class Product {
    private int productId;
    private String productName;
    private int quantity;
    private double price;

    public Product(int productId, String productName, int quantity, double price) {
        this.productId = productId;
        this.productName = productName;
        this.quantity = quantity;
        this.price = price;
    }

    public int getProductId() { return productId; }
    public String getProductName() { return productName; }
    public void setProductName(String productName) { this.productName = productName; }
```

```

    }

    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }

    public String toString() {
        return String.format("ID: %d | Name: %-15s | Qty: %4d | Price: $%.2f", produc
tId, productName, quantity, price);
    }
}

class Inventory {
    private HashMap<Integer, Product> products = new HashMap<>();

    // Add product - O(1) average
    public void addProduct(Product product) {
        if (products.containsKey(product.getProductId())) {
            System.out.println("Product ID " + product.getProductId() + " already exi
sts!");
            return;
        }
        products.put(product.getProductId(), product);
        System.out.println("Added: " + product);
    }

    // Update product - O(1) average
    public void updateProduct(int productId, String name, int quantity, double price)
    {
        Product p = products.get(productId);
        if (p == null) { System.out.println("Product ID " + productId + " not found!"
); return; }
        p.setProductName(name);
        p.setQuantity(quantity);
        p.setPrice(price);
        System.out.println("Updated: " + p);
    }

    // Delete product - O(1) average
    public void deleteProduct(int productId) {
        Product removed = products.remove(productId);
        if (removed != null) System.out.println("Deleted: " + removed);
        else System.out.println("Product ID " + productId + " not found!");
    }

    // Display all products - O(n)
    public void displayAll() {
        System.out.println("\n--- Current Inventory ---");
        if (products.isEmpty()) { System.out.println("Inventory is empty."); return;
        }

        for (Map.Entry<Integer, Product> entry : products.entrySet()) {
            System.out.println(entry.getValue());
        }
        System.out.println("-----\n");
    }
}

```

```

public class InventoryManagement {
    public static void main(String[] args) {
        System.out.println("=== Inventory Management System ===\n");
        Inventory inventory = new Inventory();
        inventory.addProduct(new Product(101, "Laptop", 50, 999.99));
        inventory.addProduct(new Product(102, "Mouse", 200, 29.99));
        inventory.addProduct(new Product(103, "Keyboard", 150, 49.99));
        inventory.addProduct(new Product(104, "Monitor", 75, 299.99));
        inventory.displayAll();
        inventory.updateProduct(102, "Wireless Mouse", 180, 39.99);
        inventory.deleteProduct(104);
        inventory.displayAll();
    }
}

```

FinancialForecasting.java

```

/*
 * Exercise 7: Financial Forecasting
 *
 * Recursion:
 * - A programming technique where a method calls itself to solve a smaller instance
of the same problem.
 * - Requires a base case to terminate recursion, and a recursive step to move toward
s the base case.
 *
 * Time Complexity of Simple Recursive Growth Calculation:
 * -  $O(N)$  where  $N$  is the number of periods/years, since we call the function once per
year.
 * - Space complexity is  $O(N)$  due to the call stack.
 *
 * Optimization:
 * - Iterative approach: reduces space complexity to  $O(1)$  by avoiding the call stack.
 * - Memoization: useful if there are overlapping subproblems (not directly applicabl
e to simple linear growth,
 *   but essential for tree-like recursive models like option pricing).
 */

```

```

public class FinancialForecasting {

    // Recursive method to calculate future value
    // Formula:  $FV = PV * (1 + growthRate)^{years}$ 
    // Recursively:  $FV(years) = FV(years - 1) * (1 + growthRate)$ 
    public static double calculateFutureValue(double currentValue, double growthRate,
int years) {
        // Base case: if years is 0, future value is the current value
        if (years == 0) {
            return currentValue;
        }
        // Recursive case
        return calculateFutureValue(currentValue, growthRate, years - 1) * (1 + growt
hRate);
    }
}

```

```

// Optimized Iterative Method (O(1) auxiliary space)
public static double calculateFutureValueIterative(double currentValue, double growthRate, int years) {
    double futureValue = currentValue;
    for (int i = 0; i < years; i++) {
        futureValue *= (1 + growthRate);
    }
    return futureValue;
}

public static void main(String[] args) {
    System.out.println("=== Financial Forecasting Tool ===\n");

    double initialInvestment = 1000.00; // Present Value (PV)
    double annualGrowthRate = 0.05;     // 5% growth rate
    int forecastingPeriod = 10;          // 10 years

    System.out.println("Initial Investment: $" + initialInvestment);
    System.out.println("Annual Growth Rate: " + (annualGrowthRate * 100) + "%");
    System.out.println("Forecast Period: " + forecastingPeriod + " years");

    // Recursive Calculation
    long startRec = System.nanoTime();
    double futureValueRec = calculateFutureValue(initialInvestment, annualGrowthRate, forecastingPeriod);
    long endRec = System.nanoTime();

    System.out.println("\n--- Recursive Approach ---");
    System.out.printf("Predicted Future Value: %.2f\n", futureValueRec);
    System.out.println("Time taken: " + (endRec - startRec) + " ns");

    // Iterative Calculation
    long startIter = System.nanoTime();
    double futureValueIter = calculateFutureValueIterative(initialInvestment, annualGrowthRate, forecastingPeriod);
    long endIter = System.nanoTime();

    System.out.println("\n--- Iterative Approach (Optimized) ---");
    System.out.printf("Predicted Future Value: %.2f\n", futureValueIter);
    System.out.println("Time taken: " + (endIter - startIter) + " ns");

    System.out.println("\n--- Analysis & Optimization ---");
    System.out.println("Recursive Complexity: Time O(N), Space O(N) due to call stack.");
    System.out.println("Iterative Complexity: Time O(N), Space O(1) - prevents StackOverflowError for large N.");
}

```

DSA Execution Log (output.txt)

```

=== Inventory Management ===
=== Inventory Management System ===

```

```

Added: ID: 101 | Name: Laptop          | Qty: 50 | Price: $999.99

```

Added: ID: 102 | Name: Mouse | Qty: 200 | Price: \$29.99
Added: ID: 103 | Name: Keyboard | Qty: 150 | Price: \$49.99
Added: ID: 104 | Name: Monitor | Qty: 75 | Price: \$299.99

--- Current Inventory ---

ID: 101 | Name: Laptop | Qty: 50 | Price: \$999.99
ID: 102 | Name: Mouse | Qty: 200 | Price: \$29.99
ID: 103 | Name: Keyboard | Qty: 150 | Price: \$49.99
ID: 104 | Name: Monitor | Qty: 75 | Price: \$299.99

Updated: ID: 102 | Name: Wireless Mouse | Qty: 180 | Price: \$39.99
Deleted: ID: 104 | Name: Monitor | Qty: 75 | Price: \$299.99

--- Current Inventory ---

ID: 101 | Name: Laptop | Qty: 50 | Price: \$999.99
ID: 102 | Name: Wireless Mouse | Qty: 180 | Price: \$39.99
ID: 103 | Name: Keyboard | Qty: 150 | Price: \$49.99

=== Ecommerce Search ===

=== E-commerce Platform Search ===

--- Linear Search ---

Found: ID: 5 | Watch | Accessories
Time: 3042 ns

Sorted Products:

ID: 4 | Book | Education
ID: 7 | Camera | Electronics
ID: 2 | Headphones | Electronics
ID: 1 | Laptop | Electronics
ID: 3 | Shirt | Clothing
ID: 6 | Shoes | Clothing
ID: 5 | Watch | Accessories

--- Binary Search ---

Found: ID: 5 | Watch | Accessories
Time: 10125 ns

--- Comparison ---

Linear Search: $O(n)$ - Best for small/unsorted datasets
Binary Search: $O(\log n)$ - Best for large/sorted datasets

=== Sorting Customer Orders ===

=== Sorting Customer Orders ===

Original Orders:

Order #1 | Alice | \$250.00
Order #2 | Bob | \$150.75
Order #3 | Charlie | \$500.50
Order #4 | Diana | \$75.25
Order #5 | Eve | \$320.00

After Bubble Sort ($O(n^2)$):

```
Order #4 | Diana      | $75.25
Order #2 | Bob        | $150.75
Order #1 | Alice       | $250.00
Order #5 | Eve        | $320.00
Order #3 | Charlie    | $500.50
Time: 2625 ns
```

```
After Quick Sort (O(n log n)):
Order #4 | Diana      | $75.25
Order #2 | Bob        | $150.75
Order #1 | Alice       | $250.00
Order #5 | Eve        | $320.00
Order #3 | Charlie    | $500.50
Time: 2500 ns
```

--- Performance Comparison ---

Bubble Sort: $O(n^2)$ - Simple but slow for large datasets

Quick Sort: $O(n \log n)$ avg - Much faster, preferred in practice

=== Employee Management ===

=== Employee Management System ===

```
Added: ID: 1 | Alice Johnson | Manager      | $70000.00
Added: ID: 2 | Bob Brown     | Developer    | $60000.00
Added: ID: 3 | Charlie Davis  | Designer     | $55000.00
Added: ID: 4 | Diana Wilson  | Analyst      | $52000.00
```

--- All Employees ---

```
ID: 1 | Alice Johnson | Manager      | $70000.00
ID: 2 | Bob Brown    | Developer    | $60000.00
ID: 3 | Charlie Davis  | Designer     | $55000.00
ID: 4 | Diana Wilson  | Analyst      | $52000.00
-----
```

Searching for Employee ID 3:

```
Found: ID: 3 | Charlie Davis | Designer     | $55000.00
Deleted: ID: 2 | Bob Brown   | Developer    | $60000.00
```

--- All Employees ---

```
ID: 1 | Alice Johnson | Manager      | $70000.00
ID: 3 | Charlie Davis | Designer     | $55000.00
ID: 4 | Diana Wilson  | Analyst      | $52000.00
-----
```

=== Task Management ===

=== Task Management System (Linked List) ===

```
Added: Task #1 | Design UI      | In Progress
Added: Task #2 | Write API       | Pending
Added: Task #3 | Unit Tests     | Pending
Added: Task #4 | Deploy        | Not Started
```

--- Task List ---

```
Task #1 | Design UI      | In Progress
Task #2 | Write API       | Pending
```

```
Task #3 | Unit Tests      | Pending
Task #4 | Deploy          | Not Started
-----
```

Searching for Task #3:

```
Found: Task #3 | Unit Tests      | Pending
Deleted: Task #2 | Write API      | Pending
```

--- Task List ---

```
Task #1 | Design UI          | In Progress
Task #3 | Unit Tests            | Pending
Task #4 | Deploy                 | Not Started
-----
```

=== Library Management ===

=== Library Management System ===

--- Linear Search ---

```
Found: ID: 3 | 1984                | George Orwell
```

Sorted Books:

```
ID: 3 | 1984                | George Orwell
ID: 6 | Moby Dick              | Herman Melville
ID: 4 | Pride and Prejudice        | Jane Austen
ID: 5 | The Catcher in the Rye     | J.D. Salinger
ID: 1 | The Great Gatsby           | F. Scott Fitzgerald
ID: 2 | To Kill a Mockingbird      | Harper Lee
```

--- Binary Search ---

```
Found: ID: 3 | 1984                | George Orwell
```

--- Comparison ---

Linear: $O(n)$ - good for small/unsorted collections

Binary: $O(\log n)$ - ideal for large sorted collections

=== Financial Forecasting ===

=== Financial Forecasting Tool ===

Initial Investment: \$1000.0

Annual Growth Rate: 5.0%

Forecast Period: 10 years

--- Recursive Approach ---

Predicted Future Value: \$1628.89

Time taken: 1417 ns

--- Iterative Approach (Optimized) ---

Predicted Future Value: \$1628.89

Time taken: 1125 ns

--- Analysis & Optimization ---

Recursive Complexity: Time $O(N)$, Space $O(N)$ due to call stack.

Iterative Complexity: Time $O(N)$, Space $O(1)$ - prevents StackOverflowError for large N

.

3. PL/SQL Programming

PL/SQL adds programming extensions to Oracle SQL, allowing variables, cursors, packages, triggers, and exception handling. We implement standard banking procedures across 7 SQL scripts:

- Exercise 1: Control Structures (Senior interest rates updates, VIP flags, loan reminders).
- Exercise 2: Error Handling (Exceptions logging on SafeTransfer, updates, additions).
- Exercise 3: Stored Procedures (Monthly interest computation, bonuses, transfers).
- Exercise 4: Functions (Loan calculations, age validation, balance check).
- Exercise 5: Triggers (Audit transactions, balances check, update timestamp).
- Exercise 6: Cursors (Monthly statements, late fee fees, interest rate tuning).
- Exercise 7: Packages (CustomerManagement, EmployeeManagement, AccountOperations).

schema.sql (Database Structure)

```
-- Schema Creation for PL/SQL Exercises

-- Drop tables if they already exist to start fresh
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE AuditLog CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE Transactions CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE Loans CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE Accounts CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE Customers CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE Employees CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/

-- Customers Table
CREATE TABLE Customers (
    CustomerID NUMBER PRIMARY KEY,
    Name VARCHAR2(100),
    DOB DATE,
    Balance NUMBER,
```

```
LastModified DATE,
IsVIP VARCHAR2(5) DEFAULT 'FALSE'
);

-- Accounts Table
CREATE TABLE Accounts (
    AccountID NUMBER PRIMARY KEY,
    CustomerID NUMBER,
    AccountType VARCHAR2(20),
    Balance NUMBER,
    LastModified DATE,
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

-- Transactions Table
CREATE TABLE Transactions (
    TransactionID NUMBER PRIMARY KEY,
    AccountID NUMBER,
    TransactionDate DATE,
    Amount NUMBER,
    TransactionType VARCHAR2(10),
    FOREIGN KEY (AccountID) REFERENCES Accounts(AccountID)
);

-- Loans Table
CREATE TABLE Loans (
    LoanID NUMBER PRIMARY KEY,
    CustomerID NUMBER,
    LoanAmount NUMBER,
    InterestRate NUMBER,
    StartDate DATE,
    EndDate DATE,
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

-- Employees Table
CREATE TABLE Employees (
    EmployeeID NUMBER PRIMARY KEY,
    Name VARCHAR2(100),
    Position VARCHAR2(50),
    Salary NUMBER,
    Department VARCHAR2(50),
    HireDate DATE
);

-- AuditLog Table (for Exercise 5 Scenario 2)
CREATE TABLE AuditLog (
    LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    TransactionID NUMBER,
    AccountID NUMBER,
    TransactionDate DATE,
    Amount NUMBER,
    TransactionType VARCHAR2(10),
    LogDate DATE DEFAULT SYSDATE
);
```

```
-- Error Log Table for logging procedures errors
CREATE TABLE ErrorLog (
    LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    ProcedureName VARCHAR2(100),
    ErrorMessage VARCHAR2(4000),
    LogDate DATE DEFAULT SYSDATE
);
```

Exercise7_Packages.sql (Modular Functions & Procedures)

```
-- Exercise 7: Packages

-- =====
-- Package 1: CustomerManagement
-- =====

CREATE OR REPLACE PACKAGE CustomerManagement AS
    PROCEDURE AddCustomer(
        p_customer_id IN NUMBER,
        p_name         IN VARCHAR2,
        p_dob          IN DATE,
        p_balance       IN NUMBER
    );

    PROCEDURE UpdateCustomer(
        p_customer_id IN NUMBER,
        p_name         IN VARCHAR2,
        p_balance       IN NUMBER
    );

    FUNCTION GetCustomerBalance(
        p_customer_id IN NUMBER
    ) RETURN NUMBER;
END CustomerManagement;
/

CREATE OR REPLACE PACKAGE BODY CustomerManagement AS
    PROCEDURE AddCustomer(
        p_customer_id IN NUMBER,
        p_name         IN VARCHAR2,
        p_dob          IN DATE,
        p_balance       IN NUMBER
    ) IS
    BEGIN
        INSERT INTO Customers (CustomerID, Name, DOB, Balance, LastModified)
        VALUES (p_customer_id, p_name, p_dob, p_balance, SYSDATE);
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Added Customer: ' || p_name || ' (ID: ' || p_customer_id || ')');
    EXCEPTION
        WHEN DUP_VAL_ON_INDEX THEN
            DBMS_OUTPUT.PUT_LINE('Error: Customer ID ' || p_customer_id || ' already exists.');
```

```

END AddCustomer;

PROCEDURE UpdateCustomer(
    p_customer_id IN NUMBER,
    p_name        IN VARCHAR2,
    p_balance     IN NUMBER
) IS
BEGIN
    UPDATE Customers
    SET Name = p_name,
        Balance = p_balance,
        LastModified = SYSDATE
    WHERE CustomerID = p_customer_id;

    IF SQL%ROWCOUNT = 0 THEN
        DBMS_OUTPUT.PUT_LINE('Error: Customer ID ' || p_customer_id || ' not found
d. ');
    ELSE
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Updated Customer details for ID: ' || p_customer_id
);
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error updating customer: ' || SQLERRM);
END UpdateCustomer;

FUNCTION GetCustomerBalance(
    p_customer_id IN NUMBER
) RETURN NUMBER IS
    v_balance NUMBER := 0;
BEGIN
    SELECT Balance INTO v_balance
    FROM Customers
    WHERE CustomerID = p_customer_id;

    RETURN v_balance;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        RETURN NULL;
    WHEN OTHERS THEN
        RETURN NULL;
END GetCustomerBalance;
END CustomerManagement;
/

```

```

-- =====
-- Package 2: EmployeeManagement
-- =====

CREATE OR REPLACE PACKAGE EmployeeManagement AS
    PROCEDURE HireEmployee(
        p_employee_id IN NUMBER,
        p_name        IN VARCHAR2,
        p_position     IN VARCHAR2,

```

```

        p_salary      IN NUMBER,
        p_department  IN VARCHAR2,
        p_hire_date   IN DATE
    );

    PROCEDURE UpdateEmployee(
        p_employee_id IN NUMBER,
        p_position     IN VARCHAR2,
        p_salary        IN NUMBER
    );

    FUNCTION CalculateAnnualSalary(
        p_employee_id IN NUMBER
    ) RETURN NUMBER;
END EmployeeManagement;
/

CREATE OR REPLACE PACKAGE BODY EmployeeManagement AS
    PROCEDURE HireEmployee(
        p_employee_id IN NUMBER,
        p_name         IN VARCHAR2,
        p_position     IN VARCHAR2,
        p_salary        IN NUMBER,
        p_department   IN VARCHAR2,
        p_hire_date    IN DATE
    ) IS
    BEGIN
        INSERT INTO Employees (EmployeeID, Name, Position, Salary, Department, HireDate)
        VALUES (p_employee_id, p_name, p_position, p_salary, p_department, p_hire_date);

        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Hired Employee: ' || p_name || ' (ID: ' || p_employee_id || ')');
    EXCEPTION
        WHEN DUP_VAL_ON_INDEX THEN
            DBMS_OUTPUT.PUT_LINE('Error: Employee ID ' || p_employee_id || ' already exists.');
```

```

        WHEN OTHERS THEN
            ROLLBACK;
            DBMS_OUTPUT.PUT_LINE('Error hiring employee: ' || SQLERRM);
    END HireEmployee;

    PROCEDURE UpdateEmployee(
        p_employee_id IN NUMBER,
        p_position     IN VARCHAR2,
        p_salary        IN NUMBER
    ) IS
    BEGIN
        UPDATE Employees
        SET Position = p_position,
            Salary = p_salary
        WHERE EmployeeID = p_employee_id;

        IF SQL%ROWCOUNT = 0 THEN
            DBMS_OUTPUT.PUT_LINE('Error: Employee ID ' || p_employee_id || ' not found');
```

```

d. ');
    ELSE
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Updated Employee ID: ' || p_employee_id);
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error updating employee: ' || SQLERRM);
END UpdateEmployee;

FUNCTION CalculateAnnualSalary(
    p_employee_id IN NUMBER
) RETURN NUMBER IS
    v_salary NUMBER := 0;
BEGIN
    SELECT Salary INTO v_salary
    FROM Employees
    WHERE EmployeeID = p_employee_id;

    -- Annual salary is Monthly Salary * 12
    RETURN v_salary * 12;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        RETURN NULL;
    WHEN OTHERS THEN
        RETURN NULL;
END CalculateAnnualSalary;
END EmployeeManagement;
/

```

```

-- =====
-- Package 3: AccountOperations
-- =====
CREATE OR REPLACE PACKAGE AccountOperations AS
    PROCEDURE OpenAccount(
        p_account_id    IN NUMBER,
        p_customer_id   IN NUMBER,
        p_account_type  IN VARCHAR2,
        p_balance       IN NUMBER
    );

    PROCEDURE CloseAccount(
        p_account_id IN NUMBER
    );

    FUNCTION GetTotalCustomerBalance(
        p_customer_id IN NUMBER
    ) RETURN NUMBER;
END AccountOperations;
/

```

```

CREATE OR REPLACE PACKAGE BODY AccountOperations AS
    PROCEDURE OpenAccount(
        p_account_id    IN NUMBER,

```

```

        p_customer_id IN NUMBER,
        p_account_type IN VARCHAR2,
        p_balance      IN NUMBER
    ) IS
        v_cust_exists NUMBER := 0;
BEGIN
    -- Check customer exists
    SELECT COUNT(*) INTO v_cust_exists FROM Customers WHERE CustomerID = p_customer_id;

    IF v_cust_exists = 0 THEN
        DBMS_OUTPUT.PUT_LINE('Error: Customer ID ' || p_customer_id || ' does not exist. Cannot open account.');
```

er_id;

```
        RETURN;
    END IF;

    INSERT INTO Accounts (AccountID, CustomerID, AccountType, Balance, LastModified)
    VALUES (p_account_id, p_customer_id, p_account_type, p_balance, SYSDATE);
    COMMIT;
    DBMS_OUTPUT.PUT_LINE('Opened Account: ' || p_account_id || ' (' || p_account_type || ') for Customer ID: ' || p_customer_id);
EXCEPTION
    WHEN DUP_VAL_ON_INDEX THEN
        DBMS_OUTPUT.PUT_LINE('Error: Account ID ' || p_account_id || ' already exists.');
```

ists.');

```
    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error opening account: ' || SQLERRM);
END OpenAccount;

PROCEDURE CloseAccount(
    p_account_id IN NUMBER
) IS
BEGIN
    DELETE FROM Accounts
    WHERE AccountID = p_account_id;

    IF SQL%ROWCOUNT = 0 THEN
        DBMS_OUTPUT.PUT_LINE('Error: Account ID ' || p_account_id || ' not found.');
```

');

```
    ELSE
        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Closed Account ID: ' || p_account_id);
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error closing account: ' || SQLERRM);
END CloseAccount;

FUNCTION GetTotalCustomerBalance(
    p_customer_id IN NUMBER
) RETURN NUMBER IS
    v_total_balance NUMBER := 0;
BEGIN
    -- Sum up balance across all accounts of the customer
```

```
        SELECT SUM(Balance) INTO v_total_balance
        FROM Accounts
        WHERE CustomerID = p_customer_id;

        RETURN NVL(v_total_balance, 0);
    EXCEPTION
        WHEN OTHERS THEN
            RETURN 0;
    END GetTotalCustomerBalance;
END AccountOperations;
/
```


4. JUnit & Mockito Testing

Test-Driven Development (TDD) builds bulletproof features by writing test cases alongside core methods. We implement a Maven Spring Boot test suite comprising 68 tests covering:

- Basic JUnit: Arrange-Act-Assert (AAA) pattern test fixtures (@BeforeEach, @AfterEach).
- Advanced JUnit: Parameterized tests (@CsvSource), exceptions assertions, method ordering, and timeouts.
- Mockito Basic: Mocking services, argument matching, interactions, verifying call sequences, void methods.
- Advanced Mockito: Stubs for Repository databases, REST API clients, file IO, network connections.
- Spring Tests: JPA entity tests using H2 databases, Web layer Controller tests with MockMvc, and SpringBootTest integration tests.

CalculatorTest.java (Basic JUnit)

```
package com.cognizant.basic;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

public class CalculatorTest {
    private Calculator calculator;

    // Test Fixture Setup
    @BeforeEach
    public void setUp() {
        System.out.println("Setting up Calculator instance before test...");
        calculator = new Calculator();
    }

    // Test Fixture Teardown
    @AfterEach
    public void tearDown() {
        System.out.println("Cleaning up Calculator instance after test...");
        calculator = null;
    }

    @Test
    public void testAdd() {
        // Arrange
        int a = 10;
        int b = 20;

        // Act
        int result = calculator.add(a, b);

        // Assert
        assertEquals(30, result, "10 + 20 should equal 30");
    }

    @Test
    public void testSubtract() {
        // Arrange
        int a = 50;
```

```
        int b = 20;

        // Act
        int result = calculator.subtract(a, b);

        // Assert
        assertEquals(30, result, "50 - 20 should equal 30");
    }

    @Test
    public void testMultiply() {
        // Arrange
        int a = 5;
        int b = 6;

        // Act
        int result = calculator.multiply(a, b);

        // Assert
        assertEquals(30, result, "5 * 6 should equal 30");
    }

    @Test
    public void testDivide() {
        // Arrange
        int a = 60;
        int b = 2;

        // Act
        double result = calculator.divide(a, b);

        // Assert
        assertEquals(30.0, result, "60 / 2 should equal 30.0");
    }

    @Test
    public void testDivideByZero() {
        // Assert exception
        assertThrows(ArithmeticException.class, () -> {
            calculator.divide(10, 0);
        }, "Dividing by zero should throw ArithmeticException");
    }
}
```

UserControllerTest.java (MockMvc Spring Controller)

```
package com.cognizant.spring;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.*;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

import java.util.NoSuchElementException;
```

```
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.http.MediaType;
import org.springframework.test.web.servlet.MockMvc;

import com.fasterxml.jackson.databind.ObjectMapper;

// Spring Test Exercise 3 & Mock Dependencies Exercise 1: Mocking Service Dependency
in a Controller Test
@WebMvcTest(UserController.class)
public class UserControllerTest {

    @Autowired
    private MockMvc mockMvc;

    @MockBean
    private UserService userService;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    public void testGetUser_Success() throws Exception {
        // Arrange
        User user = new User(1L, "Ishan");
        when(userService.getUserById(1L)).thenReturn(user);

        // Act & Assert
        mockMvc.perform(get("/users/1"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.id").value(1))
            .andExpect(jsonPath("$.name").value("Ishan"));

        verify(userService).getUserById(1L);
    }

    // Spring Test Exercise 8: Test Controller Exception Handling (GlobalExceptionHandler)
    @Test
    public void testGetUser_NotFound_Returns404() throws Exception {
        // Arrange
        when(userService.getUserById(99L)).thenThrow(new NoSuchElementException("User
not found"));

        // Act & Assert
        mockMvc.perform(get("/users/99"))
            .andExpect(status().isNotFound())
            .andExpect(content().string("User not found"));

        verify(userService).getUserById(99L);
    }

    // Spring Test Exercise 5: Test Controller POST Endpoint
    @Test
```

```

public void testCreateUser_Success() throws Exception {
    // Arrange
    User user = new User(2L, "Jane");
    when(userService.saveUser(any(User.class))).thenReturn(user);

    // Act & Assert
    mockMvc.perform(post("/users")
        .contentType(MediaType.APPLICATION_JSON)
        .content(objectMapper.writeValueAsString(user)))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.id").value(2))
        .andExpect(jsonPath("$.name").value("Jane"));

    verify(userService).saveUser(any(User.class));
}
}

```

JUnit Maven Test Execution Log (output.txt)

```

[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:Week1_JUnit_Mockito_Testing >-----
[INFO] Building Week1_JUnit_Mockito_Testing 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ Week1_JUnit_Mockito_Testing ---
[INFO] Deleting /Users/ishanparmar2105/.gemini/antigravity/scratch/cognizant/Week1_JUnit_Testing/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ Week1_JUnit_Mockito_Testing ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/ishanparmar2105/.gemini/antigravity/scratch/cognizant/Week1_JUnit_Testing/src/main/resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ Week1_JUnit_Mockito_Testing ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 14 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ Week1_JUnit_Mockito_Testing ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/ishanparmar2105/.gemini/antigravity/scratch/cognizant/Week1_JUnit_Testing/src/test/resources

```

```
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ Week1_JUnit_Mockito_Testing ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 14 source files with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ Week1_JUnit_Mockito_Testing ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running com.cognizant.advanced.OrderedTests
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.015 s -- in com.cognizant.advanced.OrderedTests
[INFO] Running com.cognizant.advanced.EvenCheckerTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.021 s -- in com.cognizant.advanced.EvenCheckerTest
[INFO] Running com.cognizant.advanced.ExceptionThrowerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s -- in com.cognizant.advanced.ExceptionThrowerTest
[INFO] Running com.cognizant.advanced.PerformanceTesterTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.613 s -- in com.cognizant.advanced.PerformanceTesterTest
[INFO] Running com.cognizant.basic.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -- in com.cognizant.basic.AssertionsTest
[INFO] Running com.cognizant.basic.CalculatorTest
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.010 s -- in com.cognizant.basic.CalculatorTest
[INFO] Running com.cognizant.mockito.advanced.AdvancedMockitoTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.431 s -- in com.cognizant.mockito.advanced.AdvancedMockitoTest
[INFO] Running com.cognizant.mockito.MyServiceTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s -- in com.cognizant.mockito.MyServiceTest
[INFO] Running com.cognizant.spring.UserControllerTest
11:36:39.527 [main] INFO org.springframework.test.context.support.AnnotationConfigCon
```

```
textLoaderUtils -- Could not detect default configuration classes for test class [com
.cognizant.spring.UserControllerTest]: UserControllerTest does not declare any static
, non-private, non-final, nested classes annotated with @Configuration.
11:36:39.571 [main] INFO org.springframework.boot.test.context.SpringBootTestContextB
ootstrapper -- Found @SpringBootConfiguration com.cognizant.spring.TestApplication fo
r test class com.cognizant.spring.UserControllerTest
```

```

      .      _      _      _      _      _      _      _      _      _      _
/\ /  _  '  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _  _
( ( ) \__ | ' _ | ' _ | | ' _ \ _ ` | \ \ \ \ \
\ \  _  ) | | _ | | | | | | | | ( _ | | ) ) ) )
'   | _  | . _ | | | _ | | _ \ , | / / / / /
=====|_|=====|_|=/ / / / /
:: Spring Boot ::                                (v3.2.0)

```

```
2026-06-24T11:36:39.686+05:30 INFO 16827 --- [main] c.cognizant.spring.UserControllerTest : Starting UserControllerTest using Java 25.0.2 with PID 16827 (started by ishanparmar2105 in /Users/ishanparmar2105/.gemini/antigravity/scratch/congnizant/Week1_JUnit_Testing)
2026-06-24T11:36:39.687+05:30 INFO 16827 --- [main] c.cognizant.spring.UserControllerTest : No active profile set, falling back to 1 default profile: "default"
2026-06-24T11:36:40.042+05:30 INFO 16827 --- [main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring TestDispatcherServlet ''
2026-06-24T11:36:40.042+05:30 INFO 16827 --- [main] o.s.t.web.servlet.TestDispatcherServlet : Initializing Servlet ''
2026-06-24T11:36:40.042+05:30 INFO 16827 --- [main] o.s.t.web.servlet.TestDispatcherServlet : Completed initialization in 0 ms
2026-06-24T11:36:40.049+05:30 INFO 16827 --- [main] c.cognizant.spring.UserControllerTest : Started UserControllerTest in 0.457 seconds (process running for 2.911)
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.637 s -- in com.cognizant.spring.UserControllerTest
[INFO] Running com.cognizant.spring.UserServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.048 s -- in com.cognizant.spring.UserServiceTest
[INFO] Running com.cognizant.spring.UserIntegrationTest
2026-06-24T11:36:40.175+05:30 INFO 16827 --- [main] t.c.s.AnnotationConfigContextLoaderUtils : Could not detect default configuration classes for test class [com.cognizant.spring.UserIntegrationTest]: UserIntegrationTest does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
2026-06-24T11:36:40.177+05:30 INFO 16827 --- [main] .b.t.c.SpringBootTestContextBootstrapper : Found @SpringBootConfiguration com.cognizant.spring.TestApplication for test class com.cognizant.spring.UserIntegrationTest
```

[illegible]

```
2026-06-24T11:36:40.185+05:30 INFO 16827 --- [          main] c.cognizant.spring.Us
erIntegrationTest : Starting UserIntegrationTest using Java 25.0.2 with PID 16827 (
started by ishanparmar2105 in /Users/ishanparmar2105/.gemini/antigravity/scratch/conq
```

```
nizant/Week1_JUnit_Testing)
2026-06-24T11:36:40.186+05:30 INFO 16827 --- [          main] c.cognizant.spring.UserIntegrationTest : No active profile set, falling back to 1 default profile: "default"
2026-06-24T11:36:40.277+05:30 INFO 16827 --- [          main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-06-24T11:36:40.288+05:30 INFO 16827 --- [          main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 8 ms. Found 1 JPA repository interface.
2026-06-24T11:36:40.372+05:30 INFO 16827 --- [          main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-06-24T11:36:40.446+05:30 INFO 16827 --- [          main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection conn0: url=jdbc:h2:mem:022ab671-39a9-4b3c-ad72-15c393381dee user=SA
2026-06-24T11:36:40.446+05:30 INFO 16827 --- [          main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-06-24T11:36:40.465+05:30 INFO 16827 --- [          main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-06-24T11:36:40.492+05:30 INFO 16827 --- [          main] org.hibernate.Version : HHH000412: Hibernate ORM core version 6.3.1.Final
2026-06-24T11:36:40.509+05:30 INFO 16827 --- [          main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-06-24T11:36:40.554+05:30 INFO 16827 --- [          main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-06-24T11:36:40.804+05:30 INFO 16827 --- [          main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-06-24T11:36:40.815+05:30 INFO 16827 --- [          main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-06-24T11:36:40.971+05:30 WARN 16827 --- [          main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-06-24T11:36:41.033+05:30 INFO 16827 --- [          main] o.s.b.t.m.w.SpringBootTestMockServletContext : Initializing Spring TestDispatcherServlet ''
2026-06-24T11:36:41.033+05:30 INFO 16827 --- [          main] o.s.t.web.servlet.TestDispatcherServlet : Initializing Servlet ''
2026-06-24T11:36:41.033+05:30 INFO 16827 --- [          main] o.s.t.web.servlet.TestDispatcherServlet : Completed initialization in 0 ms
2026-06-24T11:36:41.038+05:30 INFO 16827 --- [          main] c.cognizant.spring.UserIntegrationTest : Started UserIntegrationTest in 0.86 seconds (process running for 3.9)
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.927 s -- in com.cognizant.spring.UserIntegrationTest
[INFO] Running com.cognizant.spring.CalculatorServiceTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.011 s -- in com.cognizant.spring.CalculatorServiceTest
[INFO] Running com.cognizant.spring.UserRepositoryTest
2026-06-24T11:36:41.111+05:30 INFO 16827 --- [          main] t.c.s.AnnotationConfigContextLoaderUtils : Could not detect default configuration classes for test class [com.cognizant.spring.UserRepositoryTest]: UserRepositoryTest does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
2026-06-24T11:36:41.114+05:30 INFO 16827 --- [          main] .b.t.c.SpringBootTestContextBootstrapper : Found @SpringBootConfiguration com.cognizant.spring.TestApplication for test class com.cognizant.spring.UserRepositoryTest
```

```

.      _ _ _ _ _
/\ /  _ _ _ _ _ ( _ ) _ _ _ _ \ \ \ \
( ( ) \ _ _ | ' _ | ' _ | ' _ \ / _ _ | \ \ \ \
\ \ / _ _ ) | _ _ | | | | | | ( _ | | ) ) ) )
'   | _ _ | . _ | | | _ | | _ \ , | / / / /
=====|_|=====|_|_/ _ / _ / _ /
:: Spring Boot ::                (v3.2.0)

```

```

2026-06-24T11:36:41.121+05:30 INFO 16827 --- [          main] c.cognizant.spring.UserRepositoryTest : Starting UserRepositoryTest using Java 25.0.2 with PID 16827 (started by ishanparmar2105 in /Users/ishanparmar2105/.gemini/antigravity/scratch/cognizant/Week1_JUnit_Testing)
2026-06-24T11:36:41.121+05:30 INFO 16827 --- [          main] c.cognizant.spring.UserRepositoryTest : No active profile set, falling back to 1 default profile: "default"
2026-06-24T11:36:41.137+05:30 INFO 16827 --- [          main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-06-24T11:36:41.139+05:30 INFO 16827 --- [          main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 1 ms. Found 1 JPA repository interface.
2026-06-24T11:36:41.142+05:30 INFO 16827 --- [          main] beddedDataSourceBeanFactoryPostProcessor : Replacing 'dataSource' DataSource bean with embedded version
2026-06-24T11:36:41.150+05:30 INFO 16827 --- [          main] o.s.j.d.e.EmbeddedDatabaseFactory : Starting embedded database: url='jdbc:h2:mem:7c48e250-27b2-4a00-9d3a-96ccb7620e91;DB_CLOSE_DELAY=-1;DB_CLOSE_ON_EXIT=false', username='sa'
2026-06-24T11:36:41.158+05:30 INFO 16827 --- [          main] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-06-24T11:36:41.158+05:30 INFO 16827 --- [          main] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-06-24T11:36:41.159+05:30 INFO 16827 --- [          main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-06-24T11:36:41.167+05:30 INFO 16827 --- [          main] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
Hibernate: drop table if exists users cascade
Hibernate: create table users (id bigint not null, name varchar(255), primary key (id))
2026-06-24T11:36:41.168+05:30 INFO 16827 --- [          main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-06-24T11:36:41.181+05:30 INFO 16827 --- [          main] c.cognizant.spring.UserRepositoryTest : Started UserRepositoryTest in 0.066 seconds (process running for 4.044)
Hibernate: select u1_0.id,u1_0.name from users u1_0 where u1_0.id=?
Hibernate: select u1_0.id,u1_0.name from users u1_0 where u1_0.id=?
Hibernate: select u1_0.id,u1_0.name from users u1_0 where u1_0.id=?
Hibernate: insert into users (name,id) values (?,?)
Hibernate: insert into users (name,id) values (?,?)
Hibernate: insert into users (name,id) values (?,?)
Hibernate: select u1_0.id,u1_0.name from users u1_0 where u1_0.name=?
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.089 s -- in com.cognizant.spring.UserRepositoryTest
[INFO] Running com.cognizant.advanced.AllTests
[INFO] Running com.cognizant.basic.CalculatorTest
Setting up Calculator instance before test...

```



```
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
Setting up Calculator instance before test...
Cleaning up Calculator instance after test...
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in
com.cognizant.basic.CalculatorTest
[INFO] Running com.cognizant.basic.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in
com.cognizant.basic.AssertionsTest
[INFO] Running com.cognizant.advanced.EvenCheckerTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in
com.cognizant.advanced.EvenCheckerTest
[INFO] Running com.cognizant.advanced.ExceptionThrowerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in
com.cognizant.advanced.ExceptionThrowerTest
[INFO] Running com.cognizant.advanced.PerformanceTesterTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.613 s -- in
com.cognizant.advanced.PerformanceTesterTest
[INFO] Running com.cognizant.advanced.OrderedTests
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s -- in
com.cognizant.advanced.OrderedTests
[INFO] Tests run: 0, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.636 s -- in
com.cognizant.advanced.AllTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 68, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.862 s
[INFO] Finished at: 2026-06-24T11:36:42+05:30
[INFO] -----
```

5. SLF4J & Logback Logging

SLF4J acts as a facade pattern for multiple logging frameworks. We use Logback as the provider:

- Level Configuration: Trace, Debug, Info, Warn, Error logging.
- Parameterized Logging: Optimizing string interpolation to save memory.
- Custom Appenders: ConsoleAppender (for development stdout) and FileAppender (writing logs to app.log).

logback.xml (Appenders Configuration)

```
<configuration>
  <appender name="console" class="ch.qos.logback.core.ConsoleAppender">
    <encoder>
      <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
    </encoder>
  </appender>

  <appender name="file" class="ch.qos.logback.core.FileAppender">
    <file>app.log</file>
    <encoder>
      <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
    </encoder>
  </appender>

  <root level="debug">
    <appender-ref ref="console" />
    <appender-ref ref="file" />
  </root>
</configuration>
```

AppendersLoggingExample.java

```
package com.cognizant.logging;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class AppendersLoggingExample {
    private static final Logger logger = LoggerFactory.getLogger(AppendersLoggingExample.class);

    public static void main(String[] args) {
        System.out.println("--- Running Exercise 3: Logging to Console and File (app.log) ---");

        logger.debug("Debug log: starting process...");
        logger.info("Info log: system initialized.");
        logger.warn("Warn log: configuration is using defaults.");
        logger.error("Error log: database connection timed out, retrying...");

        System.out.println("Exercise 3 Logging completed. Check 'app.log' in the project directory.");
    }
}
```

```
}
}
```

Logging Output Log (output.txt)

=== Logging Example ===

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:Week1_SL4J_Logging >-----
[INFO] Building Week1_SL4J_Logging 1.0-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ Week1_SL4J_Logging ---
--- Running Exercise 1: SLF4J Error and Warning Levels ---
11:36:54.490 [com.cognizant.logging.LoggingExample.main()] DEBUG com.cognizant.loggin
g.LoggingExample - This is a debug message
11:36:54.490 [com.cognizant.logging.LoggingExample.main()] INFO  com.cognizant.loggin
g.LoggingExample - This is an info message
11:36:54.491 [com.cognizant.logging.LoggingExample.main()] WARN  com.cognizant.loggin
g.LoggingExample - This is a warning message
11:36:54.491 [com.cognizant.logging.LoggingExample.main()] ERROR com.cognizant.loggin
g.LoggingExample - This is an error message
Exercise 1 Logging completed successfully.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  0.212 s
[INFO] Finished at: 2026-06-24T11:36:54+05:30
[INFO] -----
```

=== Parameterized Logging ===

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:Week1_SL4J_Logging >-----
[INFO] Building Week1_SL4J_Logging 1.0-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ Week1_SL4J_Logging ---
--- Running Exercise 2: Parameterized Logging ---
11:36:55.199 [com.cognizant.logging.ParameterizedLoggingExample.main()] INFO  c.c.l.P
arameterizedLoggingExample - User 'Ishan Parmar' attempted to log in.
11:36:55.200 [com.cognizant.logging.ParameterizedLoggingExample.main()] WARN  c.c.l.P
arameterizedLoggingExample - Failed login attempt #3 for user 'Ishan Parmar' from IP
address 192.168.1.105.
11:36:55.201 [com.cognizant.logging.ParameterizedLoggingExample.main()] ERROR c.c.l.P
arameterizedLoggingExample - An error occurred during math operations for user 'Ishan
Parmar': / by zero
java.lang.ArithmeticException: / by zero
    at com.cognizant.logging.ParameterizedLoggingExample.main(ParameterizedLoggingExa
mple.java:24)
    at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodH
andleAccessor.java:104)
    at java.base/java.lang.reflect.Method.invoke(Method.java:565)
```

```
at org.codehaus.mojo.exec.AbstractExecJavaBase.executeMainMethod(AbstractExecJava
Base.java:402)
at org.codehaus.mojo.exec.ExecJavaMojo.executeMainMethod(ExecJavaMojo.java:142)
at org.codehaus.mojo.exec.AbstractExecJavaBase.doExecClassLoader(AbstractExecJava
Base.java:377)
at org.codehaus.mojo.exec.AbstractExecJavaBase.lambda$execute$0(AbstractExecJavaB
ase.java:287)
at java.base/java.lang.Thread.run(Thread.java:1474)
```

Exercise 2 Logging completed successfully.

```
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.181 s
[INFO] Finished at: 2026-06-24T11:36:55+05:30
[INFO] -----
```

=== Appenders Logging ===

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:Week1_SL4J_Logging >-----
[INFO] Building Week1_SL4J_Logging 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ Week1_SL4J_Logging ---
--- Running Exercise 3: Logging to Console and File (app.log) ---
11:36:55.920 [com.cognizant.logging.AppendersLoggingExample.main()] DEBUG c.c.logging
.AppendersLoggingExample - Debug log: starting process...
11:36:55.921 [com.cognizant.logging.AppendersLoggingExample.main()] INFO c.c.logging
.AppendersLoggingExample - Info log: system initialized.
11:36:55.921 [com.cognizant.logging.AppendersLoggingExample.main()] WARN c.c.logging
.AppendersLoggingExample - Warn log: configuration is using defaults.
11:36:55.921 [com.cognizant.logging.AppendersLoggingExample.main()] ERROR c.c.logging
.AppendersLoggingExample - Error log: database connection timed out, retrying...
Exercise 3 Logging completed. Check 'app.log' in the project directory.
```

```
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.182 s
[INFO] Finished at: 2026-06-24T11:36:55+05:30
[INFO] -----
```